Progress Report 1

Antonio Lorenzo Díaz-Meco, Pablo Gómez Martínez, Adolfo Lozano Mendoza and Mario Freire Arias

April 2024

1 Problem and Hypothesis

1.1 Describe the problem and hypotheses you have chosen to focus on. Describe how you will tackle it, at a mathematical and algorithmic level.

Our problem is Explainability Applied to RNNs. In this project, we aim to explore the different explainability methods currently available for Recurrent Neural Networks. We are exploring feature importance, variable effects and its interactions through the insights of the agnostic methods primarily, but we would also like to implement at least one specific method for RNN architectures. We are giving this project a perspective that focuses on sentiment analysis, a widely spread technique in Natural Language Proccessing.

In an algorithmic level, we have started by creating a RNN model and training it. This is a crucial step because our whole results will depend on how this model performs. We have also started to implement agnostic methods such as ICE, PDP, ALE and LIME. We have found some libraries that will help us on the implementation, but currently, we are having to implement new functions to adapt those libraries to our needs because they weren't made with sentiment analysis in mind.

For instance, instead of using Dataloaders for generating predictions we created functions that can predict and perform classification with a phrase or multiple paragraphs. This is really important to make agnostic models work. We discovered this need when we were trying to implement Lime.

We haven't tackled the theoretical and mathemathical perspective enough yet. We plan to deep-dive into this perspective soon and get familiar with all the mathemathical basis behind this models in order to have a better understanding of the theme and be able to explain the results we are finding.

In our project proposal we thought about benchmarking as the end goal of our project, but now this has changed. We are focusing primarily on understanding and implementing every model and knowing how it exactly works behind the scenes. As RNNs are very opaque due to their black-box nature. This appeared to be a more interesting topic to research. We are also looking forward to understanding every methods' advantages and disadvantages when applied to RNNs and maybe propose a solution for those problems.

2 Future

The next steps to achieve our objectives are:

1. Code implementation of agnostic methods (LIME, ALE, PDP and Permutation Importance) for the RNN model.
2. Comprehend the algorithmic intricacies and their underlying mathematical foundation.
3. Try to implement at least one specific method for RNNs (for example, XRAI or TimeSHAP).

Once these fundamental objectives are met, we would endeavour to incorporate algorithms such as SHAP or other specific algorithms. A more unrealistic objective would be to do a benchmark of all the methods to see which one is better and why.

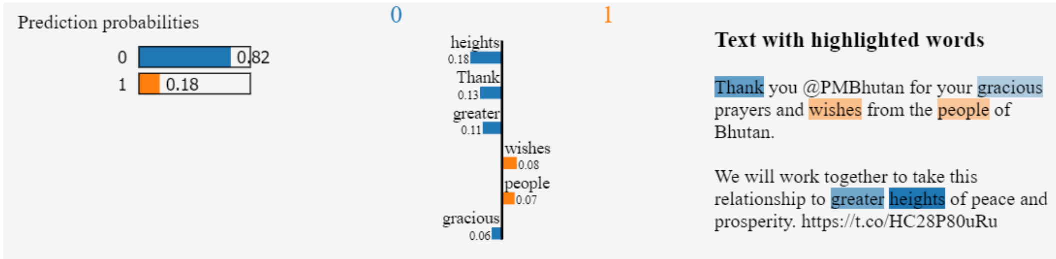

Figure 1: Lime results

3 Implementation

Initially, we utilized the data to classify texts as being generated by a human or a bot, although the ultimate goal is to achieve explainability in sentiment analysis.

Using PyTorch, we have constructed an RNN model and trained it. In addition to the necessary functions for training, we have developed a function that enables us to classify specific texts. The model consists of an initial embedding layer that converts from the indices associated to each token in the inputs to their embedding vector, followed by a RNN layer (of arbitrary dimensions, i.e., subject to change), and a fully connected layer applied to the last hidden state to generate the final output.

Taking a low-level approach to the explanatory methods will facilitate their integration into our RNN model. We have tested some of these explanatory methods on simpler models (not RNNs) to grasp their mechanisms.

Many of these methods perform well with the Sklearn library. The methods we have developed include:

1. ALE: We employed the PyAle library and adapted it for use with a RandomForestRegressor.

2. Lime: We have already implemented Lime for the human/bot classifier RNN using the LimeTextExplainer module from the lime library.

3. SHAP: By utilizing the shap library and transformers, we have successfully applied it to a Sentiment Analysis model.

4 Preliminary Results

We have successfully implemented Lime. In Figure 1 we can see the results achieved. We have obtained this using lime on a bot-human classification over a dataset of real tweets.

5 Problems

The main problem encountered is that the vast majority of examples available use categorical data, which differs from our use case of text classification. Additionally, these methods have a great performance for models developed in sklearn, making adaptation to PyTorch complex.